JAVA WITH DSA-DAY4

ARRAYS

MOVE ZEROES TO END:

```
import java.util.*;
public class Main {
    public static void main(String[] args) {
        int arr[] = {1, 0, 2, 0, 4, 0, 5};
        int index = 0;
        // Move non-zero elements forward
        for (int i = 0; i < arr.length; i++) {
            if (arr[i] != 0) {
                arr[index] = arr[i];
                index++;
            }
        }
        // Fill remaining positions with 0
        while (index < arr.length) {
            arr[index] = 0;
            index++;
        }
        for (int num : arr) {
            System.out.print(num + " ");
        }
    }
}
```

OUTPUT:

1 2 4 5 0 0 0

with two pointer

```
import java.util.*;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter n:");
        int n = sc.nextInt();
        int arr[] = new int[n];
```

```
        System.out.println("Enter elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        int left = 0;
        int right = n - 1;
        while (left < right) {
            while (left < right && arr[left]!= 0) {
                left++;
            }
            while (left < right && arr[right]== 0) {
                right--;
            }
            if (left < right) {
                int temp = arr[left];
                arr[left] = arr[right];
                arr[right] = temp;
            }
        }
        System.out.println("After moving zeroes:");
        for (int num : arr) {
            System.out.print(num + " ");
        }
    }
}
```

OUTPUT:

```
Enter n:
5
Enter elements:
1
0
5
0
6
After moving zeroes:
```

1 6 5 0 0

MAXIMUM SUM OF A SUBARRAY OF SIZE K USING SLIDING WINDOW

How it works:

Window 1 → [2 1 5] → sum = 8

Window 2 → [1 5 1] → remove 2, add 1 → sum = 7

Window 3 → [5 1 3] → remove 1, add 3 → sum = 9

Window 4 → [1 3 2] → remove 5, add 2 → sum = 6

```
class Main {
    public static void main(String[] args) {
        int arr[] = {-2, 1, -3, 4, -1, 2, 5};
        int k = 3;
        int max = 0;
        for (int i = 0; i <= arr.length - k; i++) {
            int sum = 0;
            for (int j = i; j < i + k; j++) {
                sum += arr[j];
            }
            if (sum > max) {
                max = sum;
            }
        }
        System.out.println("Maximum sum = " + max);
    }
}
```

MAXIMUM SUM OF A SUBARRAY OF SIZE K USING SLIDING WINDOW-OPTIMISED:

```
class Main {
    public static void main(String[] args) {
        int arr[] = {-2, 1, -3, 4, -1, 2, 5};
        int k = 3;
        int sum = 0;
        for(int i=0;i<k;i++){
            sum+=arr[i];
        }
        int maxx = sum;
```

```
        for (int i = k; i < arr.length; i++) {
            sum =sum-arr[i-k]+arr[i];
            maxx=Math.max(maxx, sum);
        }
        System.out.println("Maximum sum = " + maxx);
    }
}
```

PRINT THE SUM OF EVERY WINDOW:

```
class Main {
    public static void main(String[] args) {
        int arr[] = {-2, 1, -3, 4, -1, 2, 5};
        int k = 3;
        int sum = 0;
        a
        for(int i=0;i<k;i++){
            sum+=arr[i];
        }
        System.out.println(""+sum);
        int maxx = sum;
        for (int i = k; i < arr.length; i++) {
            sum =sum-arr[i-k]+arr[i];
            System.out.println(""+sum);
            //maxx=Math.max(maxx, sum);
        }
        //System.out.println("Maximum sum = " + maxx);
    }
}
```

MAXIMUM SUM OF A SUBARRAY USING SLIDING WINDOW (without size)

```
public class Main {
    public static void main(String[] args) {
        int arr[] = {-2, 1, -3, 4, -1, 2, 5, 4};
        int maxSum = arr[0];
        int currentSum = arr[0];
        for (int i = 1; i < arr.length; i++) {
            currentSum = Math.max(arr[i], currentSum + arr[i]);
```

```
            maxSum = Math.max(maxSum, currentSum);
        }
        System.out.println("Maximum subarray sum = " + maxSum);
    }
}
```

OUTPUT:

Maximum subarray sum = 14

Brute force method:

```
public class Main {
    public static void main(String[] args) {
        int arr[] = {-2, 1, -3, 4, -1, 2, 5, 4};
        int maxsum = arr[0];
        for (int i = 1; i < arr.length; i++) {
            int sum=arr[i];
            maxsum=Math.max(maxsum,sum);
            for(int j=i+1;j<arr.length;j++){
                sum+=arr[j];
                maxsum=Math.max(maxsum,sum);
            }
        }
        System.out.println("Maximum subarray sum = " + maxsum);
    }
}
```

KADANE'S ALGORITHM:

```
public class Main {
    public static void main(String[] args) {
        int arr[] = {-2, 1, -3, 4, -1, 2, 5, 4};
        int maxsum = arr[0];
        int sum=0;
        for (int i = 1; i < arr.length; i++) {
            sum+=arr[i];
            maxsum=Math.max(maxsum,sum);
            if (sum<0){
                sum==0;
            }
```

```
        }
        System.out.println("Maximum subarray sum = " + maxsum);
    }
}
```

OUTPUT:

Maximum subarray sum = 14

STRINGS

PRINT THE LENGTH OF THE LAST STRING

```
import java.util.*;
class Main {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        String a=sc.nextLine();
        String[] b=a.toLowerCase().trim().split(" ");
        System.out.println(b[b.length-1].length());
    }
}
```

OUTPUT:

HELLO WORLD

5

INPUT: C5D2A3

OUTPUT: CCCCCDDAAA

```
import java.util.*;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String s = sc.nextLine();
        for (int i = 0; i < s.length(); ) {
            char ch = s.charAt(i);
            i++;
            String num = "";
            while (i < s.length() && Character.isDigit(s.charAt(i))) {
                num += s.charAt(i);
                i++;
            }
            int count = Integer.parseInt(num);
            for (int j = 0; j < count; j++) {
                System.out.print(ch);
            }
        }
    }
}
```